CSE471: System Analysis and Design

Assignment on Functional Requirements

Proposed Project Title: Dhaka Smart Transit & Congestion Intelligence Platform

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GitHub Repository:

https://github.com/rezwanhritom/Dhaka-Smart-Transit-Congestion-Intelligence-Platform

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Project Overview

Project Description

The Dhaka Smart Transit & Congestion Intelligence Platform is a city-scale web-based system designed to provide commuters with real-time visibility into buses, routes, and congestion conditions while enabling transport authorities and operators to manage performance, incidents, and maintenance through a unified digital infrastructure.

The system integrates live bus tracking, AI-based arrival time prediction, incident reporting, congestion forecasting, and operational analytics into one secure platform to reduce uncertainty, improve punctuality, and increase transparency in Dhaka's public transport network.

This platform digitizes the complete lifecycle of the following:

- Bus trip operation and monitoring
- Commuter journey planning and alerts
- Public incident and complaint reporting
- Congestion prediction and impact estimation
- Fleet performance, SLA, and maintenance analytics

Instead of operating as separate, disconnected tools for tracking, complaints, and planning, transit data and workflows are handled within one structured, workflow-driven platform.

The system ensures:

- Centralized commuter access to live bus information
- Data-driven ETA and congestion predictions using AI models
- Structured incident and complaint management with full lifecycle tracking
- Role-based operational control for operators and transport officers
- Transparent performance metrics and public accountability dashboards
- Secure handling of GPS data, user accounts, and operational decisions

This project simulates a real-world smart transit operations center for Dhaka using proper system analysis principles, multi-service architecture, defined user roles, normalized database design, and controlled access mechanisms with integrated AI modules.

Tech Stack (MERN)

Core Web Stack

- Frontend: Next.js (React.js), HTML, CSS, TypeScript
- Backend API Gateway: Node.js (NestJS or Express.js)
- Databases: PostgreSQL (with PostGIS for geospatial), Redis (caching)
- Message Broker: Kafka or RabbitMQ (event-driven GPS and incident pipelines)

AI & Data Services

- AI Microservices: Python (FastAPI) for ETA, crowding, congestion prediction, and incident classification
- ML Models:
    - Gradient boosting / neural networks for bus arrival time prediction (ETA) using historical GPS and schedule data
    - CNN/LSTM-style models or gradient boosting for traffic congestion forecasting based on spatial-temporal traffic patterns
    - Text + metadata classifiers for incident categorization and severity scoring

Authentication & Security

- Authentication: JWT-based authentication with refresh tokens, optional Google OAuth
- Password Security: bcrypt password hashing
- Authorization: Role-Based Access Control (RBAC) following least-privilege best practices
- Rate Limiting & Security Middleware: input validation, secure headers, CSRF/XSS/SQLi protections

External APIs

- Maps API: Map provider (e.g., Mapbox / Google Maps) for map tiles and geocoding

- Email/SMS APIs: For notifications, alerts, and verification (e.g., SendGrid/Twilio)
- Weather API (optional): To enrich congestion and ETA models with weather features
- Push Notification API (optional): For mobile/web push alerts on delays and disruptions

The system follows proper REST API architecture behind an API gateway, secure middleware implementation, structured and normalized relational database schema design with geospatial support, asynchronous event processing, and controlled access using role-based access control (RBAC).

User Roles

1. Commuter
    - Register and log in
    - View nearby stops and live bus locations
    - See AI-predicted ETAs and crowding levels
    - Plan commutes and compare route options
    - Subscribe to alerts for favorite routes and trips
    - Submit incidents and complaints about buses and routes
    - Track incident status and view history of submissions
2. Bus Operator
    - Register buses and onboard GPS devices
    - View fleet performance and on-time metrics
    - Monitor active trips and route adherence
    - Receive and manage incidents related to their buses
    - Update incident status and add operational notes
    - Review maintenance hotspot suggestions for their routes
3. Transport Officer / Authority User
    - Monitor city-wide congestion map and forecasts
    - View route performance, SLA compliance, and missed trips
    - Review and manage high-severity incidents and complaints
    - Approve and track maintenance actions for problem segments
    - Access accountability dashboards and performance reports

4. System Administrator
    - Manage user accounts and role assignments
    - Configure routes, stops, and base schedules
    - Manage operator registrations and device/API key issuance
    - Monitor system statistics, health, and error rates
    - Access and review audit logs of critical actions
    - Configure backup, retention, and security policies
5. ML/DevOps Engineer (Internal Role)
    - Manage AI model training, evaluation, and deployments
    - Configure model versions and feature flags
    - Monitor model performance and system metrics
    - Coordinate rollbacks and experiments

System Characteristics

- Secure authentication & authorization with JWT, 2FA option for staff
- Role-Based Access Control across commuter, operator, officer, admin, ML/DevOps roles
- Multi-service architecture (tracking, incidents, analytics/AI) behind an API gateway
- Event-driven processing for GPS updates and incident workflows
- AI-driven ETA, congestion, crowding, and incident impact predictions using historical and real-time data
- Structured incident and complaint lifecycle tracking with full history
- Fleet performance and SLA monitoring dashboards with exportable reports
- Secure file and media upload system for incident photos/videos
- Comprehensive audit logging for every critical action and configuration change
- Powerful search and filtering across buses, routes, incidents, and time ranges
- Data export & reporting APIs for planners and researchers
- Session timeout, rate limiting, and anomaly detection on sensitive operations
- Responsive, mobile-first user interface optimized for commuters on smartphones
- Monitoring and observability stack for application performance and reliability

Target Users

- General commuters in Dhaka using public buses
- Bus operators and fleet managers
- Transport authority officials and city planners
- Traffic management and maintenance teams
- Journalists, researchers, and civic tech organizations analyzing transit data

This project represents a structured, workflow-driven, AI-enhanced full-stack transit management and accountability system aligned with software engineering principles and system analysis & design methodologies. It demonstrates layered/microservice architecture, secure RBAC, real-time data processing, and practical application of machine learning in a realistic smart city context.

Functional Requirements

SL	Common Workflows
1	User registration & login system Commuters, operators, officers, and admins register/log in with email/phone; passwords are hashed, sessions are securely managed, and roles are assigned.
2	Bus device registration & authentication Operators register buses and obtain API keys or tokens; devices use these to authenticate GPS and telemetry requests to the tracking service.

Module 1	
Member	Feature Description
1	AI-powered commute planner Users enter origin/destination and desired arrival time; system suggests best route(s), departure time, and bus options, factoring in congestion predictions and crowding estimates.

2	Nearby stops & live bus discovery system Commuters can search or use location to view nearby stops, see upcoming buses, live locations, and AI-predicted arrival times for each route at each stop.
3	Personalized alerts & notifications system Commuters can subscribe to routes/trips and receive real-time notifications for major delays, cancellations, or severe congestion affecting their usual commute.

Module 2	
Member	Feature Description
1	Incident classification & routing system The system automatically classifies incidents using rules + AI model, assigns severity, and routes them to the appropriate operator or authority team based on route/zone.
1	AI-based impact estimation system For high-severity incidents, AI estimates affected routes and likely delay duration and suggests reroutes; the system updates ETAs and alerts commuters automatically.
2	Citizen incident & complaint submission system Commuters submit structured incidents (e.g., breakdowns, unsafe driving, overcrowding, and road blockages) with category, bus/route, GPS location, description, and photo/video upload.
2	Public transparency & accountability dashboard The public view shows anonymized, aggregated incidents; average resolution times by operator/zone; and open issues per area, improving transparency and accountability.

3	Incident lifecycle tracking system Officers and operators manage incidents through stages (Reported → Verified → In Progress → Resolved → Archived), with every status change, remark, and timestamp recorded.
3	Maintenance hotspot & priority ranking system The system aggregates historical incidents and delays to identify recurring problem segments, ranks them by impact (affected passengers/time lost), and generates monthly maintenance priority lists.

Module 3	
Member	Feature Description
1	Real-time congestion monitoring & prediction system The transport authority sees a city map with current congestion and AI-predicted congestion for upcoming time windows, enabling proactive management (e.g., adding buses, altering routes).
1	Route configuration & optimization system Admins manage routes, stops, and schedules; the system suggests potential optimizations based on demand distribution and congestion patterns.
1	AI model training & configuration management system Authorized ML/DevOps staff can retrain ETA/congestion models on new data, deploy new versions, roll back models, and configure feature flags for experiments.

2	Fleet performance & SLA monitoring system Operators and officers view dashboards with metrics like on-time percentage, average delays per route, missed trips, and SLA compliance with filters by time range and route.
2	System health & observability dashboard Admins and DevOps teams see API latency, error rates, queue backlog, ingestion throughput, and storage usage, with alerts for abnormal behavior.
2	Data export & integration API system Provides secure, rate-limited APIs and export functions so external research or policy tools can access aggregated, anonymized transit and congestion data.
3	Access control & role management system Admins create, update, and deactivate user accounts; assign roles and permissions; and control which dashboards and actions each role can access.
3	Security & audit log review system Admins can filter audit logs by user, action, endpoint, and time; all sensitive operations (permission changes, device registrations, incident closures) are auditable.
3	Backup, retention & disaster recovery management system Scheduled backups, configurable data retention per data type, and documented restore procedures to ensure availability and integrity of critical data.